



EGYPT

COUNTRY REPORT ON GENOME EDITING LANDSCAPE

ANALYSIS ON AGRICULTURE

May 2026

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LIST OF ABBREVIATIONS AND ACRONYMS

ARC	Agricultural Research Centre
AGERI	Agricultural Genetic Engineering Research Institute
ASRT	Academy of Scientific Research and Technology
ABSP	Agricultural Biotechnology Support Project
ACSAD	Arab Centre for the Studies of Arid Zones and Drylands
ABS	Access and Benefit Sharing
AOF	Alliance of Science
BSL2	Biosafety level 2
CASP	Central Administration for Seed Production
CASC	Central Administration for Seed Testing and Certification
CRISPR	Clustered Regularly Interspaced Short Palindromic Repeats
CGIAR	Consultative Group on International Agricultural Research
CBD	Convention on Biological Diversity
Cas9	CRISPR-associated protein 9
CUFA	Cairo University Faculty of Agriculture
DNA	Deoxyribonucleic Acid
DRC	Desert Research Centre
EBIC	Egypt Biotechnology Information Centre
EEAA	Egyptian Environmental Affairs Agency
EPA	Environmental Protection Agency
IBC	Institutional Biosafety Committee
IPR	Intellectual Property Right
GETI	Gene Editing Technology Initiative

GMO	Genetically Modified Organism
GE _d	Genome editing
HCST	Higher Council for Science and Technology
HGP	Human Genome Project
ICARDA	International Centre for Agricultural Research in the Dry Areas
IITA	International Institute of Tropical Agriculture
IPR	Intellectual Property Right
ISAAA	International Service for the Acquisition of Agri-biotech Applications
MALR	Ministry of Agriculture and Land Reclamation
MOE	Ministry of Environment
MOH	Ministry of Health
MOHESR	Ministry of Higher Education and State for Scientific Research.
NBF	National Biosafety Framework
NFSA	National Food Safety Authority
NRC	National Research Centre
NBNE	National Biotechnology Network of Expertise
NA-RISP	North Africa Regional Integration Strategy Paper
PCT	Patent Cooperation Treaty
PCR	Polymerase Chain Reaction
PPP	Public-Private Partnership
RNA	Ribonucleic Acid
RNAi	RNA interference
STDF	Science and Technology Development Fund
STI	Science, Technology, and Innovation
SDN	Side-directed nuclease

SCU	Supreme Council of Universities
SLU	Swedish University of Agricultural Sciences
UAWC	Union of Agricultural Work Committees
UNEP	United Nations Environment Programme
UNFAA	United Nations Framework Convention on Climate Change
USDA	United States Department of Agriculture
USAID	United States Agency for International Development
WIPO	World Intellectual Property Organization

EXECUTIVE SUMMARY

Background

Genome editing (GE) in Egypt has evolved through leading institutions, including the Faculty of Agriculture, Cairo University (CUFA), and the Agricultural Genetic Engineering Research Institute (AGERI), primarily in agriculture and, more recently, in health through the Human Research Project (HRP). Egypt has a regulatory framework (currently facing operational challenges) for biotech and GE products. The Egyptian Environmental Affairs Agency, Ministry of State for Environmental Affairs, is the primary authority for overseeing biosafety and biotechnology. At the same time, the National Food Safety Authority serves as a Participating Institution/Agency. The Biosafety Law was drafted to provide a legal framework for regulating genetically modified organisms (GMOs). It was used from 2008 to 2012 to approve insect-resistant maize. Regarding GE, Egypt has not yet established specific regulations or guidelines.

Despite this, the country has a handful of GE research projects, conducted primarily at universities and research institutions, including CUFA and the Agricultural Research Centre (ARC). GE projects were initiated in 2018, with a primary focus on crops. To date, there are no GE projects for livestock, agroforestry, or fisheries. Genome editing is being clinically evaluated for the treatment of inherited diseases such as sickle cell disease and cystic fibrosis.

Regarding funding, the Science and Technology Development Fund (STDF) and the Academy of Scientific Research and Technology (ASRT) are the leading national funding agencies for research projects, and they already support and fund GE projects. Private-sector involvement is essential; however, several obstacles impede it in Egypt, including the absence of a regulatory framework for genome-edited products, weak public-private collaboration, limited awareness, and barriers to funding and incentives.

Egypt, like many African countries, faces significant global challenges related to climate change, population growth, food security, and poverty. Advances in GE technologies could be among the solutions for crop improvement, climate change mitigation, and adaptation. GE, such as CRISPR/Cas genome editing, enables precise nucleotide-level modification comparable to that achieved by natural recombination in conventional breeding. Although Egypt began research on GE in 2018, there are no policies or guidelines for GE to facilitate its use, despite its having signed and ratified international agreements, such as the Cartagena Protocol on Biosafety (2000) and the Nagoya Protocol (2012).

In a bid to fully comprehend the technology landscape, the African Union Development Agency–New Partnership for Africa's Development (AUDA-NEPAD) commissioned a Genome Editing

(GE) Landscape Analysis for Egypt to assess the country's readiness to develop, deploy, and commercialize genome-edited products. Consequently, the country's policy, regulatory, institutional, infrastructural, and technical capacities for modern biotechnology and GE were assessed to identify key enablers and constraints across scientific, socioeconomic, cultural, and governance dimensions. Specifically, the analysis examined opportunities for GE to address priority national needs at scale, including agricultural productivity, climate adaptation, food and nutrition security, post-harvest loss reduction, and diversification of healthy diets. It also identified staple and indigenous crops with high potential to improve livelihoods.

The assessment drew on secondary data (published literature and institutional databases) and primary data collected through interviews, surveys, and questionnaires. Findings were synthesized across areas such as international treaty commitments, regulatory and policy frameworks, ongoing GE projects and programmes, human and institutional capacity, research and training infrastructure, priority crops and traits, intellectual property considerations, and funding mechanisms.

Key findings

Egypt is among the first African countries to recognize the importance of cultivating biotech crops. It cultivated insect-resistant maize (2008-2012), but this was halted due to limited public awareness and political will. The rapid development of genome-editing technologies, particularly CRISPR/Cas9, has opened new avenues for fundamental research in plant biotechnology and created significant opportunities for crop genetic improvement and agricultural applications. CUFA and AGERI are leading genome-editing initiatives that use the CRISPR/Cas system in wheat, sorghum, maize, potato, and chickpea. However, as noted above, the country lacks a regulatory framework to oversee the development and deployment of these products.

This report explores the country's current GE research, development, and regulatory framework, including projects across research institutions and universities. It further highlights the infrastructure, equipment, and human capital available in universities and research institutions. The report also identifies indigenous and commercial crops and traits that can be improved using GE technology. The role of the private sector in GE technology development is also highlighted. The constraints that limit the widespread adoption of GE technologies in Egypt, including regulatory development, infrastructure investment, capacity building, and stakeholder engagement, are also identified. The review concludes by examining funding opportunities for GE research and development in the country and providing recommendations on next steps.

The following is a brief overview of the status of GEd in the country:

- GEd-related research is going on in Egypt, mainly in cereals, legumes, and indigenous crops such as cotton and potato.
- The main key drivers for GEd developments are AGERI, CUFA, DRC, ICARDA, and NRC.
- Training on GEd and regulatory literacy is limited, although the academic base in molecular biology, genetics, and bioinformatics is well practiced.
- The country lacks a specific regulatory framework for genome editing. A draft biosafety law addressing GMOs was prepared and is pending Parliament's approval, leaving GEd oversight undefined.
- Private-sector involvement remains limited and is mostly absent from current GEd-related initiatives.
- Priority and indigenous crops with strong potential for GEd applications include wheat, barley, maize, chickpea, and faba bean.
- The central funding agencies in Egypt for GEd projects are STDF and ASRT.
- The Egyptian Authority for Intellectual Property covers patents, copyrights, and plant variety protection, aligned with international treaties.
- Marketing of GE seed varieties requires approval and registration. The Seed Registration Committee handles this.
- The National Food Safety Authority (NFSA) regulates imports of modified foods (including GMOs and GEd)

Policy implications

There is an urgent need to establish a science-based national regulatory framework in Egypt to oversee the development of GEd crops as a first step toward commercialization. In addition, the National Biosafety Committee (NBC) should be activated by ratifying the biosafety law through the Egyptian Parliament. The designated areas for GMO field trials are already available for testing GEd crops. Several countries do not classify transgene-free plants produced by small indels, base-pair changes, and specific short-sequence changes as genetically modified organisms (GMOs) and therefore are exempt from GMO regulation. Egypt should consider doing the same. To secure public acceptance, additional media and stakeholder engagements are required to accelerate the country's adoption of GEd crops, thereby mitigating climate change, strengthening food security, and addressing population growth.

The following interventions should be undertaken on both short- and long-term bases to enable the country to benefit from the technology sustainably.

Short-term interventions (12–18 months)

- Provide training to key stakeholders on international and African GEd guidelines and policies from existing frameworks.
- Establishing interim GEd regulatory guidelines with a comprehensive legal framework on international best practice, clearly distinguishing genome editing from transgenic GMOs.
- Enact the Biosafety Law to activate the National Biosafety Committee, which contains an inter-ministerial committee (agriculture, research, economic planning, finance, health, environment, trade, IP).
- Establish the GEd Biosafety Committee independent from the National Biosafety Committee.
- Increase national funding for GEd projects (particularly for Indigenous and Staple Crops), to sustain current research momentum and expand training.
- Conduct stakeholder and public engagement campaigns to raise awareness about elucidating genome editing and reducing resistance.
- Build a database of ongoing research on GEd, thesis, training, equipped laboratories, etc.

Medium to long term (2–5 years)

- Ratify a comprehensive genome editing regulatory framework by enacting a dedicated GEd law, setting up risk assessment protocols, and harmonizing regulations with regional and international partners.
- Increase infrastructure and training investment to expand the number of universities and research institutes with requisite facilities for genome editing.
- Motivate public–private partnerships and build a trustworthy relationship to offer sustainable financial support for GEd projects and training.
- Extend the collaboration projects and training to cover the regional area, as well as with other African countries.
- Leveraging from developed GEd by licensing at the plant variety protection of Intellectual Property Rights.

Conclusion

Research on genome editing in Egypt is growing rapidly, but the main obstacle remains the lack of an adequate regulatory framework and limited infrastructure. It is essential to present clear, enabling policies and guidelines; targeted investment in people and infrastructure; and improved data coordination to deliver the message that the country is ready for GEd investment in GEd to tackle the climate emergency, food, and nutrition insecurity.

1. INTRODUCTION

1.1. Description of Egypt's Agricultural Landscape

Agriculture is a key sector of the Egyptian economy. Egypt's agricultural landscape is defined by the fertile Nile Valley and Delta, which account for most cultivable land and are almost entirely dependent on Nile River irrigation. The landscape includes "old" irrigated lands and newly reclaimed desert areas, a narrow rain-fed coastal strip, and diverse farming practices. According to the World Bank, Egypt's agricultural sector contributed approximately 13.7% of GDP in 2024 and employed approximately 18.7% of the total workforce. Key crops are grown in three seasons: winter wheat and clover; summer cotton and rice; and extensive vegetable cultivation. The agricultural landscape is changing due to climate change, economic factors, and population growth, making it a prime market. Agricultural research in Egypt is primarily focused on crop improvement and variety development, with an emphasis on optimizing returns and addressing biotic and abiotic stresses. With limited arable land and water for agricultural production, agricultural biotechnology offers several avenues to enhance crop development and improve resilience to climate change, including greater tolerance to drought, heat, and salinity.

1.2. National Frameworks and Investments

In Egypt, scientists view modern biotechnological techniques as promising tools for crop improvement and for novel applications of plants, animals, and microorganisms. However, there are concerns about the safety of the modified crops and animals produced by these technologies for human health and the environment. Robust national biosafety regulatory systems are required to ensure the safe use of biotechnology products. Establishing a biosafety review system entails multiple facets and associated challenges. Defining national guidelines will require investment in personnel responsible for their implementation and management. Unlike conventional products, agricultural biotechnology products use novel gene combinations and/or expression modifications to confer altered, often unique, phenotypes.

In 2008, Egypt was the first Arab country to commercialize GM corn. However, in 2012, a ministerial decree was issued suspending all commercial cultivation of GE crops. The country remains one of the most advanced developing countries in the adoption and use of agricultural biotechnology. A handful of successful gene-editing research projects, including the use of the CRISPR/Cas tool, have been carried out at the ARC and the Faculty of Agriculture at Cairo University to improve drought tolerance in Egyptian wheat cultivars and chickpeas, as well as to enhance biomass quality in sorghum (to improve its value as a forage and biomass crop).

Egypt's Central Administration for Seed Testing and Certification (CASC), established in 1995, is the agency responsible for seed quality control, seed legislation, and policy enforcement. The

Central Administration for Seed Production (CASP) administers and advises ARC on requirements for foundation and registered classes of seeds and plants. The agency supervises and contracts with seed growers to multiply seed. 66 registered seed production companies specialize primarily in hybrid maize, sorghum, Sudan grass forages, sunflower, some vegetable seeds, and clover and alfalfa seeds. In addition, 53 companies are registered for seed imports and 148 for seed exports.

1.3. Engagement in Regional Frameworks

Egypt is strengthening its position in biotechnology through national strategies and engagement in regional frameworks, with a particular focus on agricultural research and green technologies. The country is leveraging international partnerships to build capacity and infrastructure, notably through EU-Mediterranean initiatives and engagement with the Africa CDC. Egypt is also a key partner in the Partnership for Research and Innovation in the Mediterranean Area (PRIMA), with its commitment extended through 2027. This programme focuses on developing sustainable food systems and water management through biotechnology. Egypt also formally joined the Horizon Europe programme in October 2025, enabling researchers to participate in EU-funded biotechnology and environmental projects. Although research is advanced, Egypt still lacks a ratified, comprehensive biosafety law to regulate large-scale field trials and the commercialization of agricultural GMO and GEd products.

1.4. Challenges and Opportunities Justification of GEd Assessment based on Current Issues Identified in Egypt

Egypt has been implementing GEd since 2018; however, key challenges include the absence of a GEd-specific regulatory system, the need to navigate diverse global regulatory frameworks, securing public acceptance, addressing potential health and environmental risks, and resolving ethical concerns, such as equitable access to technology and corporate monopolization. Opportunities in agricultural genome editing include enhanced food security through increased crop yields, disease resistance, and adaptation to climate change, as well as improved nutritional content and sustainability.

1.4.1. Challenges Justifying GEd Assessment

Implementing gene editing (CRISPR-Cas) in Egypt faces significant challenges, primarily driven by high costs, limited infrastructure in rural areas, and ethical, religious, or cultural hesitation. Technical obstacles include off-target effects and delivery methods, compounded by a lack of comprehensive national regulations and specialized expertise for advanced applications.

Key Challenges include:

- **Economic and Resource Constraints:** GEd researched requires substantial resources and specialized human capacity, both of which remain limited in Egypt. In addition, the high cost of genetic and genomic services makes widespread application difficult.

- **Ethical and Religious Considerations:** There is significant hesitation regarding the ethical implications of gene editing, especially concerning germline modification, with some cultural/religious perspectives viewing certain technologies as unacceptable or against religious, particularly Islamic, teachings.
- **Technical and Safety Hurdles:** Technical limitations, such as off-target effects (unintended DNA cuts), mosaicism, and inefficient delivery systems (e.g., in vivo), pose significant safety risks, particularly in human applications.
- **Infrastructure and Data Disparities:** There is a scarcity of Egyptian-specific genomic and variant databases, leading to a reliance on Western data which may not reflect the local population's genetic makeup.
- **Regulatory Gaps:** Challenges exist in implementing comprehensive national biosafety laws to regulate new genome editing technologies.
- **Public Awareness and Acceptance:** Limited awareness among the public and the need for better education on the nuances of gene editing versus traditional medicine.

1.4.2. Opportunities Supporting GEd Investment

Egypt's agricultural sector is rapidly integrating gene editing (CRISPR-Cas9) to bolster food security, improve climate resilience, and increase crop yields. Investment opportunities are heavily concentrated in developing drought-tolerant, high-yield, and disease-resistant seeds, particularly for strategic crops like wheat and maize

Key Investment:

- **Climate-Resilient Seed Development:** High demand exists for breeding crops that can withstand salinity and heat stress, given Egypt's arid climate.
- **Strategic Crop Enhancement:** Investment is directed at improving staples like wheat and maize to reduce reliance on imports.
- **Biotech Infrastructure & R&D:** Different research institutes and universities are actively involved in R&D, and opportunities exist for private investment in local greenhouse facilities and specialized laboratories.
- **Agricultural Start-ups:** There is a growing focus on supporting early-stage Agri-tech firms focusing on sustainable land use and smart farming systems

1.5. Objectives of the Landscape Analysis

The overall objective of the Genome Editing (GEd) Landscape Analysis is to provide an in-depth assessment of existing policies, infrastructure, institutions, and technical capabilities for product development and commercialization across 16 African countries. For Egypt, the objective of this assessment is to:

- Provide an evidence-based analysis of Egypt's current GEd landscape, highlight policy and institutional gaps, and identify opportunities to integrate genome editing into national agricultural and health strategies.
- Identify the emerging needs in Egypt, where GEd and staple and indigenous crops require rapid improvement, focusing on agricultural productivity, reducing postharvest losses, adapting to climate change, ensuring food and nutrition security, and promoting diversified and healthy diets.
- Inform policymakers, researchers, and stakeholders about the state of GEd research and regulation in Egypt and propose actionable recommendations for establishing a coherent and functional innovation ecosystem.
- Provide an in-depth assessment and analysis of existing policies, infrastructure, institutional, and technical capabilities to support product development and commercialization in a select number of African countries.

2. APPROACH/METHODOLOGY

Data were collected from both primary and secondary sources by three researchers and the country champion. Primary data were collected through questionnaires, live interviews, an online data collection kit (ODK), surveys, and email correspondence. Leading universities engaged in GEd and/or plant biotechnology (e.g., Cairo University, Ain Shams University, Alexandria University) and research institutes (e.g., ARC, NRS, DRS, ICARDA) were visited to verify existing projects. Face-to-face interviews were conducted using questionnaires to collect data on their infrastructure and ongoing research. In addition, online interviews were conducted with contacts from universities outside Cairo. Stakeholder components derived from primary and secondary data were collected, synthesized, and presented in tables to provide a perspective on Egypt's preparedness for the adoption of GEd technologies. Secondary data were obtained from the published literature, databases, and websites. The primary data further confirmed and/or strengthened the secondary data.

Components of the regulatory and policy frameworks were identified and documented using secondary data (published literature and institutional websites) and primary data (interviews, surveys, and email correspondence). The researchers reviewed key policy documents, including the national agricultural and biotechnology strategies, the draft National Biosafety Bill, and the National Food Safety Authority (NFSA) (<https://www.elwatannews.com/news/details/820153>). They also analysed published scientific literature from Egyptian research institutions, including CUFA, AGERI, DRC, NRC, and Ain Shams University. Reference was made to the AUDA-NEPAD Consolidated Synthesis Report for cross-country benchmarking.

Data from these two sources (primary and secondary) were then analysed, synthesized, and packaged under the following thematic areas:

2.1. Regulatory and Policy Frameworks for Biotechnology and Genome Editing

Regulatory and policy frameworks, biotech, particularly GEd projects, crops, and traits, key stakeholders (partnerships) involved, and sources of funding were documented using secondary data (published literature and website databases) and primary data collection. Completed and ongoing projects, as well as published research on GEd, were analysed alongside stakeholder survey data.

2.2. Genome Editing Projects

The synthesis documented CRISPR-based research projects (in tomato, wheat, and barley), traits under development, and funding sources (institutional and donor-supported). This provided a perspective on the need to implement GEd to enhance socio-economic potential and climate

resilience. Information on biotechnology and genome editing projects, particularly genome editing initiatives, was compiled through secondary and primary data sources. This included documentation of target crops, livestock, fisheries, and forestry species, traits under development, key stakeholders and partnerships involved, and funding sources. The synthesized data were used to:

- i. identify emerging economic, social, and environmental/climate-related needs; and
- ii. assess the status of human capacity and infrastructure supporting genome editing technologies in the target countries.

2.3. Traditional and Staple Crops For Genome Editing Improvement

The collected data on GEd projects from the interviews were further disaggregated (categorized) in terms of those with the highest potential that need GEd technology for national socio-economic impact and the possibility of successful completion in view of national acceptance, resource requirements, and scalability.

2.4. Institutional Capacity (Human Capital, Infrastructure and Equipment)

Data on existing institutional capacities, including human capital, lab and field infrastructure, and equipment for GEd R&D, commercialization, and scaling, were collected from respondents during the primary data collection.

2.5. Stakeholder Mapping

Key stakeholders where appropriate, personnel were identified to provide critical data on existing biotech and GEd technologies' interventions, spread across the five stakeholder categories identified in the data collection tools (including the questionnaires, namely, regulatory agencies, research organizations/institutions, universities, private sector/industry and other government agencies/ministries and policymakers.

Stakeholders were categorized into five (5) groups, as reflected in the data collection tools:

- Regulatory agencies
- Research organizations and institutions
- Universities
- Private sector and industry
- Government ministries, agencies, and policymakers

2.6. Database Systems and Data Management

Appropriate data collection tools and platforms to support primary data collection were developed. The questionnaires were tailored to identify and map stakeholder categories, including regulatory bodies, research institutions, universities, the private sector, government agencies, and other

agencies. They were used to produce datasets (data systems). The datasets assessed Egypt's preparedness (capabilities) to adopt and scale up GEd technologies fully. To initiate primary data collection, enumerators were identified and trained virtually on ODK and the associated data-collection questionnaires/tools, delivered by the IT team and the train-the-trainers (ToT) from the Africa Harvest and AGTECH consortium. The questionnaires and ODK platforms were pretested before use. Both methodological and data triangulation were used to reduce bias and enhance the validity and reliability of the findings. All collected data were encrypted and stored in a secure, centralized server, ensuring participants' confidentiality and compliance with ethical standards.

2.7. Data Synthesis and Data Analysis

Data collected was synthesized and packaged into appropriate information, and where applicable, tables were generated.

Variables investigated included, but were not limited to:

- i. signatory to multinational food and environmental agreements;
- ii. regulatory framework components;
- iii. applications received and approved (field testing, registration, and commercialization);
- iv. R&D institutions, personnel, and laboratory/field facilities;
- v. projects;
- vi. crops and livestock species; and
- vii. traits under GEd development.
- viii.

2.8. Interactive Map

An interactive map, comparable to the Agenda 2063 Dashboard, was developed to visualize genome editing activities and capacities across Africa and is accessible at: <https://genome.africa/>

3. RESULTS

3.1. Membership in Multilateral Environmental Treaties and Agreements

Egypt is a party to several major international environmental and food agreements and actively engages in global efforts to manage biodiversity, food safety, and climate change (Table 1). Egypt has been a member of the Codex Alimentarius Commission (CAC) since 1972. The Egyptian Organization for Standardization and Quality (EOS) serves as the National Codex Contact Point, and the Egyptian Codex Committee regulates food safety and quality standards in line with international standards.

Egypt ratified the UNFCCC in 1994, the Kyoto Protocol in 2005, and the Paris Agreement in 2017. It hosted COP27 in Sharm El-Sheikh in 2022. Egypt is a party to the CBD and hosted the 14th Meeting of the Parties (COP14) in 2018, during which it served as president until 2021. Egypt is a party to the Cartagena Protocol on Biosafety, which regulates the movement of living modified organisms (LMOs). The Egyptian Environmental Affairs Agency (EEAA) manages the Egyptian Biosafety Clearinghouse (Eg-BCH) to facilitate implementation of this protocol. It is also a party to the Nagoya Protocol on Access to Genetic Resources and the Benefit-sharing (ABS) of the Convention on Biological Diversity (CBD).

Table 1: Status of Egypt's participation in key multilateral environmental agreements

Multilateral Environmental Agreements) / Treaties	Date of Ratification/ Accession by Egypt	Reference(s)
Codex Alimentarius Commission is a joint body of FAO and WHO established to develop international food standards, guidelines, and codes of practice.	Member since 1963	https://www.fao.org/fao-who-codexalimentarius/about-codex/members/en/
Convention on Biological Diversity (CBD)	Ratified 1994	https://www.cbd.int/doc/legal/cbd-en.pdf
EPA , Cartagena Protocol on Biosafety	Ratified 2003	https://bch.cbd.int/protocol/background#:~:text=The%20Cartagena%20Protocol%20on%20Biosafety,force%20on%2011%20September%202003.
EPA , Nagoya Protocol on Access and Benefit Sharing (ABS)	Ratified 2014	https://www.cbd.int/abs/nagoya-protocol/signatories
UNFCCC United Nations Framework Convention on Climate Change	Ratified 1992	https://unfccc.int/process/the-convention/status-of-ratificationhttps://unfccc.int/process-and-meetings/the-convention/status-of-ratification-of-the-convention. "

Kyoto Protocol part of its broader engagement with climate change initiatives, having also ratified the UNFCCC	Ratified 2005	https://unfccc.int/node/61057
CASC The Central Administration for Seed Testing and Certification	1995	https://www.seedtest.org/en/membership/central-administration-for.html

3.2. Policies that Advance or Hinder Adoption of Modern Biotech and GEd

Sustainable Agricultural Development Strategy updated from Vision 2030: The government supports agricultural innovation through initiatives investing in research centres. These frameworks prioritize boosting strategic crop yields, enhancing food security, and deploying targeted research and technology transfer via the Ministry of Planning, Economic Development and International. The 2030 Updated Sustainable Agriculture Development Strategy sets ambitious targets to increase agriculture's share of GDP and employment, running against the development trajectory observed in other countries, and to boost production, yields, and self-sufficiency rates for strategic crops. Responsibilities for agricultural and food policies in Egypt fall across different ministries and agencies, creating challenges for policy co-ordination and implementation. The Ministry of Agriculture and Land Reclamation, the Ministry of Supply and Internal Trade, and The Ministry of Water Resources and Irrigation all oversee some agricultural policies. Other ministries and agencies play important roles related to trade, rural development or the environment.

The patent system in Egypt, overseen by the Egyptian Patent Office, allows protection for biotechnological products and processes, including gene-related inventions.

The lack of a ratified biosafety law prevents the country from moving beyond laboratory research to large-scale, open-field trials and commercialization of home-grown biotech solutions. The government has historically demonstrated interest in the potential benefits of biotech for food security but moves cautiously, as evidenced by the lack of updated, functional regulatory frameworks.

National Food Safety Authority (NFSA): As of January 2026, the NFSA is implementing Decision No. 1/2025 (published in 2026), which defines GM foods and mandates strict handling, creating a crucial transitional period where industry actors must adapt to increased scrutiny. To overcome these hurdles, there are efforts to advance training on GEd regulation for key stakeholders (scientists, farmers, media) in 2026 to foster "science-based" approaches.

3.3. National Regulatory Framework

Egypt has developed a National Biosafety Framework (NBF) and a GMO biosafety law, and has commercialized Bt maize. However, the law was banned in 2011 due to restrictions on exporting products to the European market. Egypt ratified the Cartagena Protocol on Biosafety (2000), the Convention on Biological Diversity (1992), and the Nagoya Protocol (2013). An updated draft biosafety law was reviewed and submitted to the Egyptian Parliament, pending final approval. The National Food Safety Authority (NFSA) regulates imports of GM-derived food and feed but does not regulate domestic genome-editing research. NFSA is the key body regulating Genetically Modified Organisms (GMOs) under Law 1/2017, responsible for safety assessments, setting rules for GM foods, and aligning with Codex standards. However, a unified, legally binding biosafety framework remains lacking, leading to reliance on decrees and inter-ministerial coordination for imports and research, with the NFSA focusing heavily on risk management for both imported feed/food and locally cultivated products.

However, agricultural biotechnology faces challenges regarding the need for a comprehensive national biosafety law to replace the current reliance on various ministerial decrees, limited public awareness and trust in biotechnology and responsibilities for policy implementation are split among multiple institutions, causing coordination issues.

Egypt permits the import of agricultural biotech products if they are authorized in the country of origin, particularly for animal feed. Egypt has suspended planting and commercial release of GE crops since 2012. While research continues, regulatory gaps, including a dormant National Biosafety Committee, limit field trials and commercialization

3.3.1. Biosafety Act/Law, Biosafety Regulations/Decrees and GEd Guidelines

Egypt's efforts to address environmental responsibility for biotechnology products began in 1992 through a collaboration between the Agriculture and Genetic Engineering Research Institute (AGERI) and the Agricultural Biotechnology Support Project (ABSP). Egypt's national biosafety system was formally established by the Ministry of Agriculture and Land Reclamation (MALR) through two decrees issued in early 1995. Ministerial Decree No. 85 (January 25, 1995) establishes the National Biosafety Committee (NBC), -and Ministerial Decree No. 136 (February 7, 1995) adopts biosafety regulations and guidelines for Egypt. The first biosafety regulations and guidelines were published in draft form in January 1994 (MARL, 1994). The guidelines call for the establishment of an Institutional Biosafety Committee (IBC) at all institutions conducting recombinant DNA research.

Egypt is a Party to the Convention on Biological Diversity (CBD) and has ratified its primary instruments, including the Cartagena Protocol on Biosafety and the Nagoya Protocol. In 2008, Egypt was the first Arab country to commercialize Genetically Engineered (GE) corn in the region. However, in 2012, Egypt issued a ministerial decree suspending all commercial cultivation of GE crops. The suspension was largely driven by media campaigns against GM products and the lack of a comprehensive biosafety law.

Despite the moratorium, agricultural biotechnology research capacity has evolved and expanded through a network of universities and national research institutions. Although Egypt lacks biotechnology legislation, the government permits biotech imports when the country of origin also consumes these products (Table 2).

Egypt has commercialized *Bt*-maize, and the steps for its release and approval are presented in Figures 1 & 2. The main components of the National Biosafety System are the National Biosafety Committee (NBC), the Institutional Biosafety Committee (IBC), and the Biosafety Guidelines. Examples of biosafety milestones and provisions in Egypt are summarized in Table 2.

The Egyptian biosafety framework involved key ministries, which operate under the umbrella of the National Biosafety Committee to ensure safety in biotechnology and laboratory practices. These ministries include the Ministry of Agriculture and Land Reclamation (MALR), the Ministry of Environment (MOE), the Ministry of Higher Education and State for Scientific Research (MOHESR), and the Ministry of Health (MOH).

The GE guidelines for Egypt are currently unavailable. For GMOs, Ministerial Decree 85 established the National Biosafety Committee, which is responsible for setting regulations and policies on the safe use of genetic engineering and molecular biology to ensure environmental and human health safety. Ministerial Decree No. 136 requires and facilitates the development of GE products and mandates that a permit from the National Biosafety Committee be obtained before their use for experimental purposes. Decree 1648 states that the registration and marketing of GE seed varieties require approval by the Seed Registration Committee, which receives guidance from the National Biosafety Committee. The Egyptian Environmental Affairs Agency (EEAA), under the umbrella of the Ministry of Environment, established a biosafety unit in 2013 to assess the impact of releasing GE crops into the environment. There are three main components for issuing guidelines and granting permission to release products/food:

Environmental safety

- Contained use, experimental releases, products

- NBC, Decrees 85, 136

Food Safety

- Products intended for food
- Supreme Council on Food Safety, Decree 242

Variety registration

- Seed for cultivation
- Variety Committee - after OK National Biosafety Committee (NBC) and National Food Safety Authority (NFSA).

Table 2: Regulatory and institutional landscape for genome editing in Egypt

Institutions	Mandate / Relevance to GEd	Regulatory instruments	Date of enactment or publication	Coverage/ scope	Reference(s)
Prime Minister	Regulations governing Biotechnology	Ministerial Decree No 85	1995	Setting regulations and guidelines concerning the safe use of genetic engineering and molecular biology	https://documents1.worldbank.org/curated/en/195041468762553195/pdf/260280White0cover0BIOSAFETY1ESW.pdf
Prime Minister	GMO regulation	Ministerial Decree No 136	1995	Permission from the National Biosafety Committee is required before using or dealing with any GE.	https://cgspace.cgiar.org/server/api/core/bitstreams/882c015c-8757-46e7-9fe1-2d10acb93c5d/content
Prime Minister	R&D, Commercialization, and Trade	Decree 1648	1998	The Seed Registration Committee grants permission to register genetically modified seeds, and marketing GE seed varieties requires approval.	https://apps.fas.usda.gov/newgainapi/api/Report/DownloadReportByFileName?fileName=Biotechnology%20and%20Other%20New%20Production%20Technologies%20Annual_Cairo_Egypt_EG2023-0021
Prime Minister	GMO regulation	Decree 19	2007	Nominated new members to the National Biosafety Committee	https://www.cbd.int/doc/world/eg/eg-nbsap-powpa-en.pdf
Prime Minister's office	Genome Editing Guidelines	None	None	None	

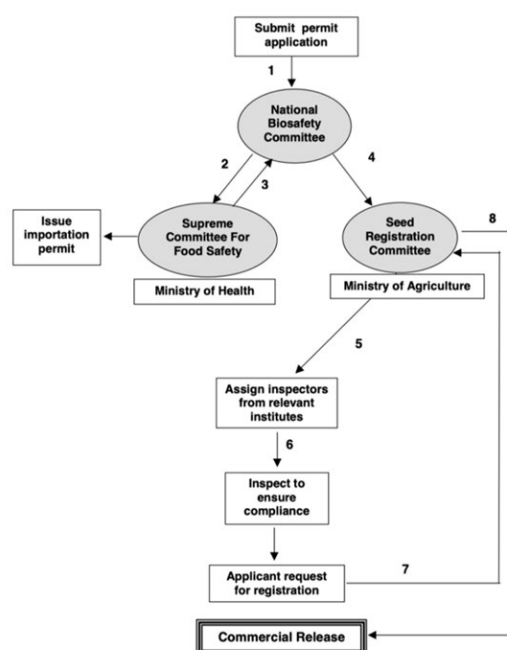
3.3.2. Regulations and Guidelines

Procedures for commercializing GMO crops were established in 1998 by Ministerial Decree No. 1648. The sequence of steps and interactions among government agencies are diagrammed in the flow chart on Figure 1 and summarized on Figure 2.

For varieties produced within Egypt, the process is as follows:

- The applicant completes a permit application form providing details of the genetic material introduced, and provides data from food and feed safety studies and evidence supporting a determination of low or negligible environmental risk.
- The application form is submitted to the NBC, which, after examination and approval, forwards it to the Seed Registration Committee for their preliminary approval to proceed with standard field trials conducted at several locations. The SRC assigns a team of qualified inspectors drawn from relevant ARC units and/or private certified laboratories to supervise cultivation, ensure adherence to any biosafety requirements, confirm the new phenotype, and evaluate agronomic performance.
- The NBC has the right to confirm the nature of the genetic modification by taking samples from the field for molecular analysis.
- After successful completion of the field trials and submission of a report to the NBC, the NBC authorizes the applicant to submit an application to the Seed Registration Committee for final approval to commercially release the new variety. Pending this, three years or seasons of agronomic performance trials are conducted under the supervision of the SRC.

Figure 1: Steps for commercial release of genetically engineered plants in Egypt (Madkour et al., 2000).



Although Egypt has some experience in assessing biotech crops, the regulatory landscape has historically been slow and precautionary. New 2026 regulations from the National Food Safety Authority (NFSA) impose full-chain control over genetically modified (GM) foods, requiring strict compliance regarding, transport, import, and sale.

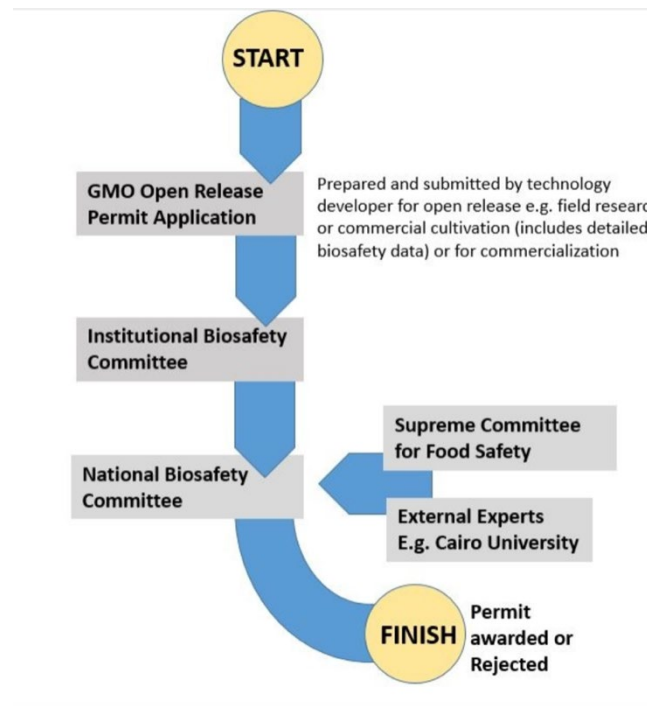


Figure 2: Regulatory Review and Approval Process (Abdallah 2010)

3.4. Socio-Economic Consideration for Decision-making in GEd Application

The integration of socio-economic considerations is an essential step toward the good governance of modern biotechnology. This can be accomplished through sound research and regulatory processes, as well as meaningful public engagement. A lack of information and analysis on the social and economic impacts of modern biotechnology affects decisions. Thus, data collection will provide a starting point for informing national governments and other stakeholders in designing and implementing policies and mechanisms that incorporate socio-economic considerations into decision-making.

Analysis of the social and economic impacts highlighted the following output:

- GEd could potentially improve crop yields and lead to higher incomes for farmers
- The regulatory framework for GEd should be considered to govern the development and application of GEd technologies and to fulfil market access and trade requirements.
- GEd communication and advocacy campaigns should be conducted to foster public awareness and acceptance to build public trust.

- Target GEd research addressing solutions for nutritional deficiencies and diseases to benefit the Egyptian people.
- The GEd guidelines and policy framework approved by other African countries, such as Kenya and Nigeria, could serve as the basis for developing Egypt's GEd guidelines.
- Training and education are required to build local capacity for GEd research, development, and regulation, which are essential.
- Essential infrastructure needs to be provided for research institutions to enable researchers to complete GEd projects and to open the door to national and international collaboration on best practices in GEd.
- Ensure alignment of the developed GEd crops with EU regulations to avoid export rejections.

3.5. Factors that Hinder or Advance Application of Modern Biotech and GEd

The application of modern biotechnology and genome editing (GEd) in Egypt—particularly in agriculture—is hindered by a complex interplay of regulatory caution, economic constraints, public scepticism, and cultural-religious sensitivities. While research exists, commercialization faces significant barriers. Factors hindering this technology include:

Political factors:

- Biosafety Concerns: There is a strong focus on precautionary safety measures, which can lead to lengthy approval processes

Geopolitical and trade factors

- Dependence on EU Regulations: Egypt is a major exporter of agricultural products. Fear that European markets will shun imports from countries using GM varieties is a significant deterrent to commercializing GEd crops.
- Strict Labelling and Traceability: New EU regulations and Egyptian policy focus on traceability—requiring physical separation of GM from non-GM crops—which adds high logistical costs.
- Geopolitical Competition: Biotech is increasingly seen through a lens of national security, which can complicate international collaboration and technology transfer

Socio-economic factors

- Smallholder Vulnerability: A large portion of Egyptian agriculture relies on smallholder farmers who may lack the capital to purchase expensive, patented GE seeds, creating risks for technology adoption.
- Input Costs: High costs of technologies and the need for specialized machinery hinder the widespread adoption of advanced farming techniques.
- Limited Infrastructure: Inadequate infrastructure for handling specialized seed multiplication and quality control can prevent new technology from reaching the field.

Cultural and religious considerations

- **Religious and Ethical Views:** In Islamic thought, while scientific advancement is encouraged, there is a strong emphasis on stewardship and "natural order." Genetic engineering is generally permitted for health and agricultural necessity (to alleviate hunger), but it faces scrutiny regarding "playing God" or altering creation unnecessarily.
- **Sanctity of Lineage (Human GEd):** Muslim scholars generally prohibit altering human reproductive cells (germline editing) because it could disturb lines of lineage.
- **"Tampering with Nature":** Cultural and religious views on the "essence" of living organisms lead to hesitancy regarding the manipulation of organisms, which is often viewed as disrespecting the natural, divine order.

Social perceptions and human factors

- **Public Scepticism and Misinformation:** Studies indicate that a significant majority of the public in the MENAT region believes GMOs are harmful to health and the environment, often driven by a lack of knowledge and fear of the unknown.
- **Misinformation:** There is a high level of fear surrounding "designer babies" or unforeseen long-term consequences of editing human genes.
- **Low Institutional Trust:** A lack of trust in regulatory bodies can exacerbate scepticism about the safety of new technologies.

3.6. Genome Editing Programmes and Projects

Egypt has an emerging but uneven capacity for genome editing. AGERI and CUFA offer training workshops and support the implementation of regional CRISPR projects in cereal, legume, and other essential crops. Several research institutes in Egypt continue to face shortages of skilled researchers in GEd. According to the Supreme Council of Universities (SCU), Egypt has 30 public universities, 19 private universities, 34 joint universities with international partners, and 18 government universities. The Agricultural Research Centre has 16 research institutes. The National Research Centre comprises 14 institutes and 109 departments across the main areas of industry, health, environment, agriculture, basic sciences, and engineering. The Desert Research Centre has 4 divisions and 11 stations. Building strong institutional capacity and fostering skilled human capital are essential to developing robust biotechnology policies and effectively absorbing and adapting modern technologies in Egypt. These universities offer a few degrees dedicated solely to GEd; however, they provide relevant courses and research opportunities within undergraduate and graduate programmes in Molecular Biology and bioinformatics that include GEd components. Two PhD students have obtained their degrees in GEd from Ain Shams University. At Cairo University, one PhD student is about to complete her degree, and two MSc students have completed their degrees in GEd. A handful of successful gene-editing research projects have been conducted primarily at universities and research institutions (Table 3).

Table 3: Genome editing projects, organisms (Crops/Forestry/Livestock/Fisheries), and traits in Egypt

Project #1: Developing Abiotic Stress Tolerant Wheat Using Gene Editing [Completed] Principal Investigators Partnership: Faculty of Agriculture, Cairo University, and the United States Agency for International Development (USAID) Nature of partnership: Joint research-(Public) Funding Source: Science and Technology Development Fund (STDF) and USAID							
Project Duration (Years)	Organism	Trait(s)	GE Technology	Amount (USD million)	Target country/countries	Development stage (DS):	Reference(s)
(2018-2022)	<i>Triticum aestivum</i> (Bread wheat)	Drought tolerance	CRISPR/Cas9 SDN1	0.4	Egypt-USA	Research & Development (R&D)	Abdallah et al., 2021
Project #2: Genome Editing in Sorghum for Improving Biofuel Production and Forage Quality. Partnership: AGERI and University of Florida Nature of Partnership: Joint research (Public) Funding Source: STDF and USAID							
(2018-2023)	Sorghum	Biofuel Production and Forage Quality	CRISPR-Cas9 SDN1	0.4	Egypt-USA	Research & Development (R&D)	Assem et al., 2023
Project #3: Developing Drought-Tolerant Chickpea Using Gene Editing Partnership: Faculty of Agriculture, Cairo University, AGERI, ICARDA, and Tuskegee University Nature of Partnership: Joint research (Public) Funding Source: STDF and USAID							

2024-2027	Chickpea	Drought tolerance	CRISPR-Cas9 Base editing SDN1	0.4	Egypt-USA	Research and Development (R&D)	Abdallah et al., 2025
Project #4: Improving maize productivity under water scarcity using precise genome editing Partnership: Faculty of Agriculture, Cairo University, AGERI, International Centre for Agricultural Research in the Dry Areas (ICARDA) Nature of Partnership: Funding Source: STDF							
2024	Maize	Drought tolerance and improving productivity	CRISPR-Cas9 Base editing, Gene activation SDN1	0.2	Egypt	R&D	Reference (in process) PI: Prof. Naglaa Abdallah
Project #5: Decreasing of Monosaccharide Contents Resulted from Cold- Induced Sweetening in Tetraploid Potato (<i>Solanum tuberosum</i> L.) Using Genome Editing. Partnership: AGERI and Ain Shams University Nature of Partnership: Joined research Funding Source: Self-funding							
2023	Potato	Cold- Induced Sweetening	CRISPR/Cas9 SDN1	PhD	Egypt	R&D	Hassan et al., 2023
Project #6: Improved starch quality by reducing the levels of amylose, thus increasing the amylopectin content. Partnership: AGERI and Ain Shams University Nature of Partnership: Joint research (Public) Funding Source: Self-funding							
2023	Potato	Starch	CRISPR/Cas9	PhD	Egypt	R&D	Ali et al., 2023

		quality	SDN1				
Project #7: CRISPR/Cas9-mediated mutagenesis of FT/TFL1 in petunia improves plant architecture and early flowering Partnership: Date Palm Research and Development, Agricultural Research Centre, and Faculty of Agriculture, Ondokuz Mayis University, Turkey. Nature of Partnership: Funding Source: Ondokuz Mayis University							
2024	Petunia	Compact flowers, smaller plants with high flower abundance	CRISPR/Cas9 SDN1	??	Egypt – Turkey	R&D	Abdulla et al. (2024) Secgin et al., 2022
Project #8: CRISPR/Cas9 Mediated Knockout of the OsbHLH024 Transcription Factor Improves Salt Stress Resistance in Rice Partnership: Taif University, Alexandria University Nature of Partnership: Public Funding Source: Taif University Researchers Supporting Project							
2022	Rice	Salt tolerance	CRISPR/Cas9	?	Saudi Arab, Egypt	R&D	Alam et al., 2022

3.7. Human Capital and Institutional Capacity

Egypt possesses strong research capabilities in biotech within public institutions. While Egypt is a pioneer in biotechnology research in the region, the lack of a clear, facilitating regulatory framework prevents these innovations from moving from laboratory to farm. According to the Supreme Council of Universities (SCU) (<https://scu.eg/universities-institution/>), Egypt has 30 public universities, 19 Egyptian Private Universities, 34 joined with International Universities, 18 Governmental Universities.

Agricultural Research Centre has sixteen research Institutes (<http://www.arc.sci.eg/ARCIstituteList.aspx?TabId=1&lang=en>), the National Research Centre consists of 14 institutes and 109 departments covering the main areas of industry, health, environment, agriculture, basic sciences and engineering (<https://www.nrc.sci.eg/about-us>), Desert Research Centre has 4 division and 11 stations (<https://drc.gov.eg/en/history/>). Building strong institutional capacity and fostering skilled human capital are essential to developing robust biotechnology policies and effectively absorbing and adapting modern technologies in Egypt.

3.7.1. Women, Youth and special Interest Groups Involvement in Modern Biotech and GEd

Women, youth, and special interest groups in Egypt are actively involved in modern biotechnology, driven by national strategies (Vision 2030) that emphasize sustainable development, agricultural resilience, and economic empowerment.

Women represent make up a significant portion of the scientific workforce (44% of researchers in scientific fields being female), with high representation in life sciences. Programmes like the L'Oréal-UNESCO For Women in Science Egypt Young Talents programme promote young female scientists in biotechnology, genetics, and molecular biology. Initiatives such as the Newton-Mosharafa Fund and UN Women's "Rabeha" programme support women in developing biotechnology startups and in gaining leadership positions in science and technology. Women still face pressures that make them underrepresented in top leadership positions, prompting specialized programmes.

Young researchers are involved in Agri-biotech ventures through initiatives like Bioteel (youth-led climate-smart farming) and incubators offering pre-incubation programmes for agribusiness and biotechnology. Different universities and research institutes offer specialized training and direct industry mentorship. for youth in genome editing and molecular biology.

Several non-profit organizations facilitating networking, training, and connecting young talents with decision-makers in the biotechnology, such as EAGEBT (Egyptian Association for Genetic Engineering & Biotechnology).

3.7.2. Research, Development and Academic Institutions

There are several R&D and academic institutions in Egypt that are engaged in various activities of GEd technologies. Examples include:

- Faculty of Agriculture, Cairo University, Ain Shams University, involved in GEd R&D, education and research training of BSc, MSc and PhD students through curricula
- National Research Institutes, Agriculture Research Institute and Desert Research Institute involved in advanced biotechnology research and field-testing
- NBNE involved in Biotechnology and GEd communication, and extension services

3.7.3. Training and Professional Development

Fundamental application of Gene editing and its regulation

The workshop and training held on 26-20 April 2026 was organized by the National Biotechnology Network of Expertise (NBNE) (one of the Academy of Scientific Research and Technology Networks), with support from ICARDA AGERI, ICARDA-lead Consortium of Red Palm Weevil Control (C4RPWC) Programme contributed to the high-level plenary session on the first day and AGERI will host the rest of training programme at the venue. The workshop will cater to a diverse group of stakeholders, including farmers, media professionals, scientists, and decision-makers, with the aim of fostering cross-sectoral dialogue and collaboration. The event was divided into two parts:

- Plenary session (26 April 2026): 100 participants
- Training (27-30 April 2026): 30 young scientists from different Universities and research institutes attended the workshop.

Hands-on experience was the core component of the training, including bioinformatics for encompassing guide RNA design, vector selection, vector construction, Biolistic and Agrobacterium-mediated plant transformation, identification of mutants, and molecular characterization of mutant lines, among other aspects. Videos for both plenary session and training were uploaded at NBNE YouTube channel.

North African, Regional Webinars on Gene Editing Technology Initiative (GETI)

A three-day webinar on gene-editing for sustainable agriculture in Africa was organized by the Network of African Science Academies (NASAC) and Africa Harvest from September 10 to 12, 2022. The webinar included 7 lectures: The role of researchers in advancing gene-editing for sustainable agriculture in Africa (Dr. Shireen Assem, ARC, Egypt); Education of

researchers for gene-editing research (Prof. Abdelfattah Badr, Helwan University); Plant genome engineering: a biotechnology tool for crop improvement (Prof. Moemen Hanafy, NRC); Agriculture genome editing in health care. Benefits and challenges (Prof. Hany Khalil, Sadat City University); The role of regulators in advancing gene editing for sustainable agriculture in Africa (Prof. Naglaa Abdallah, Cairo University); Current situation and regulations of North African countries, risks, and safety considerations (Dr. Hesham Elshishtawy, AGERI); Gene editing: from science to business (Dr. Ehab Abdel Rauf Essawy, Helwan University).

Science communication workshop on genome editing and GMO

In 2023, the National Biotechnology Network of Expertise (NBNE) and the Alliance for Science co-hosted a workshop on effective communication and advocacy strategies to raise awareness and foster a broader understanding of modern biotechnology, particularly genome editing in Egypt. The workshop engaged key stakeholder groups in developing targeted communication and advocacy strategies to support policies and decision-making processes that could enhance the uptake of genome editing and GMOs to optimize agriculture in the country in a sustainable manner.

Training Course “CRISPR/Cas9 genome editing: a new approach for therapeutics (hands-on course)”

The training, partially funded by the International Centre for Genetic Engineering and Biotechnology, covered the Basic Principles of Genome Editing Technologies: From ZNF to CRISPR; Different modified CRISPR/Cas9 approaches and Non-Cas9 Systems for Gene Editing; CRISPR vs. RNAi Technologies: History, Strengths and Weaknesses, Opportunities, Challenges and Future Prospects; and Ethical issues related to genome editing. Participants included PhD students who are involved in or plan to work in the field of gene editing, postdocs with previous experience in gene editing and who are currently working or plan to work in the same field, and academics from various fields who are interested in including gene editing in their future projects.

Use of Modern Breeding Techniques in Agriculture to form Arab Biotech Network

In 2016, the Arab Centre for the Studies of Arid Zones and Drylands (ACSAD) and the Egypt Biotechnology Information Centre (EBIC) co-hosted a workshop on biotechnology applications for developing drought-tolerant crops at the Desert Research Centre. Participants came from Libya, Jordan, Syria, Algeria, Palestine, Kuwait, Iraq, Tunisia, and Sudan. They shared and learnt more about the status of modern biotechnology in the Arab countries. An Arab Biotechnology Network was established under the umbrella of ACSAD.

YouTube Channel training on Genome editing

Gene editing is a set of genetic engineering techniques that rewrite an organism's genetic material. It is more accurate than earlier genetic engineering methods and is used to treat many plant, animal, and human diseases. Prof. Dr. Naglaa Abdullah, Professor of Agriculture in Cairo and Coordinator of the National Network of Biotechnology Experts, uses YouTube to offer online tutorials on GEd. Summarized training programmes on functional genomics and genome editing are presented in Table 4.

Table 4: Overview of functional genomics and genome editing-related training programmes in Egypt

Title of the Training Programme	Institution / Organization	Target Audience / # of Trainees per annual	Duration/Date	Gaps Identified
Fundamental application of Gene editing and its regulation	NBNE, AGERI, ICARDA, IITA	Plenary session (90 participants from stakeholders, regulators, Private sectors, scientists, media Training (30) young scientists	23-30 April 2026	Needs to be repeated to reach more participants.
North African, Regional Webinars on Gene Editing Technology Initiative (GETI)	NASAC	Scientists, stakeholders	10 - 12, September 2022	Needs more announcement to reach more people. Longer training is requested
Science communication workshop on genome editing and GMO	NBNE, Alliance for Science	Plenary session (100 participants from stakeholders, regulators, Private sectors, scientists, media Training (35)	23-26 October 2023	Needs to be repeated to reach more participants.
Training Course "CRISPR/Cas9 genome editing: a new approach for therapeutics (hands-on course)	ICGEB, Faculty of Pharmacy, Helwan University	Young researcher, PhD holders	21-23 September 2022	Needs to be repeated to reach more participants
Use of Modern Breeding Techniques in Agriculture	ACSAD, NBNE, EBIC	50 Scientists from different Arab countries	10-14 October 2016	Needs to be repeated to reach more participants

University courses

GE_d topics are covered in several courses, including molecular genetics, gene transfer, and bioinformatics, in BSc programmes at the Faculty of Agriculture, Cairo University (which has three biotechnology programmes at the Faculty of Agriculture, the Faculty of Science, and the Faculty of Veterinary Medicine), and at Ain Shams University, through undergraduate (levels 3 and 4) as well as graduate courses (MSc and PhD). The practical components also include hands-on techniques for GE_d. A summary of research institutes and universities that offer biotechnology programmes is presented in Table 5.

Table 5: Universities offering biotechnology programmes and GEd-related courses

University	BSc		Master		PhD	
	Programme	Courses	Programme	Courses	Programme	Courses
Cairo University Faculty of Agriculture	Biotechnology (English) Biotechnology (Arabic)	General Genetics Introduction to Biotechnology Molecular Genetics 1 Molecular Genetics 2 Bioinformatics Gene transfer	Genetics & Biotechnology	Advanced Genetics Biochemical Genetics Environmental Biotechnology	Genetics & Biotechnology	Bioinformatics Molecular Genetics Omics, Environmental biotechnology.
Ain Shama University Faculty of Agriculture	General Biotechnology	General Genetics Introduction to Biotechnology Molecular Genetics Bioinformatics	Genetics	Advanced Genetics Biochemical Genetics Environmental Biotechnology	Genetics	Bioinformatics Molecular Genetics Environmental Biotechnology
American University in Egypt	---		Biotechnology	Genetics Biotechnology Molecular Genetics Bioinformatics		
Alexandria University	Biotechnology	General Genetics Introduction to Biotechnology Molecular Genetics Bioinformatics	Genetics	Advanced Genetics Biochemical Genetics Environmental Biotechnology	Genetics	Bioinformatics Molecular Genetics
Mansoura University	Agriculture Biotechnology	General Genetics Introduction to Biotechnology	Genetics	Advanced Genetics	Genetics	Genetics

University	BSc		Master		PhD	
	Programme	Courses	Programme	Courses	Programme	Courses
		Molecular Genetics bioinformatics		Biochemical Genetics Environmental Biotechnology		
Benha University	Agriculture Biotechnology	General Genetics Introduction to Biotechnology Molecular Genetics Bioinformatics	Genetics	Advanced Genetics Biochemical Genetics Environmental Biotechnology		
Badr University	Biotechnology	General Genetics Introduction to Biotechnology Molecular Genetics Bioinformatics	No MSc programme		No PhD programme	
Misr University for Science and Technology (MUST)	Biotechnology	General Genetics Introduction to Biotechnology Molecular Genetics Bioinformatics Industrial Biotechnology, Environmental Biotechnology, Health Biotechnology	Biotechnology	Advanced Genetics Biochemical Genetics Environmental Biotechnology	No PhD programme	
Galala University	Molecular Biotechnology	General Genetics Introduction to Biotechnology Molecular Genetics Bioinformatics	No MSc programme		No PhD programme	

University	BSc		Master		PhD	
	Programme	Courses	Programme	Courses	Programme	Courses
October University for Modern Sciences and Arts (MSA)	Biotechnology Focus on Agricultural Biotechnology, Health Biotechnology, Environmental Biotechnology, Industrial Biotechnology	General Genetics Introduction to Biotechnology Molecular Genetics Bioinformatics Industrial Biotechnology, Environmental Biotechnology, Health Biotechnology	No MSc programme		No PhD programme	
University for Modern Sciences & Arts	Biotechnology	General Genetics Introduction to Biotechnology Molecular Genetics Bioinformatics Industrial Biotechnology, Environmental Biotechnology, Health Biotechnology	Biotechnology	Advanced Genetics Biochemical Genetics Environmental Biotechnology	No PhD programme	
German University in Cairo	Pharmacy and Biotechnology Faculty	Genetics & Genetic Engineering II Bioinformatics Introduction to Biosafety Genetics & Genetic Engineering I	Biotechnology Bioinformatics		Biotechnology	

3.8. Infrastructure and Equipment

3.8.1. Universities

The selected Egyptian universities and research institutions have the necessary infrastructure and equipment to conduct GEd-related projects. Examples of universities with the human capacity to train and conduct GEd-related projects are presented in Table 6. The Faculty of Agriculture at Cairo University has the necessary equipment and faculty to develop biotech crops.

Table 6: Status and Needs Assessment of Biosafety Laboratory Facilities by Universities

Institution	Type of Facility	Biosafety Level	Status	Limitations	Support Needed
Cairo University, Faculty of Agriculture	Research Park with all molecular biotechnology facilities for Molecular Bioinformatics, Gene transfer, tissue culture. GMO lab, Genetic Engineering lab, etc.	BSL 2	Fully equipped	Limited national funding agency Source for equipment manganous Funding novel equipment Need Bio contained greenhouse. Sequencer	Continuous funds are required, facilitate importing fine chemicals and equipment,
Ain Shama University, Faculty of Agriculture	Molecular lab, tissue culture, Bioinformatics	BSL 2	Partially equipped	Limited national funding agency Source for equipment manganous Funding novel equipment	Continuous funds are required, facilitate importing fine chemicals and equipment,
Alexandria University	Molecular lab, tissue culture,	None	Not-fully equipped	Limited national funding agency Source for equipment manganous Funding novel equipment	Continuous funds are required, facilitate importing fine chemicals and equipment,
Mansoura University	Molecular lab, tissue culture,	None	Not-fully equipped	Limited national funding agency	Continuous funds are required,

	glasshouses, Bioinformatics			Source for equipment manganous Funding novel equipment	facilitate importing fine chemicals and equipment,
October University for Modern Sciences and Arts (MSA)	Molecular lab, tissue culture, Bioinformatics	None	Partially equipped	Limited national funding agency Source for equipment manganous Funding novel equipment	Continuous funds are required, facilitate importing fine chemicals and equipment,
Misr University for Science and Technology (MUST)	Molecular lab, tissue culture, Bioinformatics	None	Partially equipped	Limited national funding agency Source for equipment manganous Funding novel equipment	Continuous funds are required, facilitate importing fine chemicals and equipment,
Nile University	Bioinformatics	None	Not-fully equipped	Limited national funding agency Source for equipment manganous Funding novel equipment	Continuous funds are required, facilitate importing fine chemicals and equipment,
Benha University	Molecular lab, tissue culture, glasshouse	None	Not-fully equipped	Limited national funding agency Source for equipment manganous Funding novel equipment	Continuous funds are required, facilitate importing fine chemicals and equipment,

3.8.2. Research Institutions

Examples of research institutes with the human capacity to train and conduct GEd-related projects are presented in Table 7. AGERI is a scientific Institute based at ARC with all the equipment and facilities needed for developing biotech crops.

Table 7: Status and needs assessment of biosafety laboratory facilities by research institutions

Institution	Type of Facility	Biosafety Level	Status (see Annex A)	Limitations	Support Needed
Agricultural Genetic Engineering Research Institute (AGERI)	Molecular lab with PCR machines, glasshouses	BSL 2	Fully equipped	Funding gap due to limited funding agency	Specialized procurement/ waiver/ exemptions, maintenance
National Research Centre (NRC)	Molecular lab with PCR machines, glasshouses	BSL 2	Fully equipped	Funding gap due to limited funding agency	Specialized procurement/ waiver/ exemptions, maintenance
Desert Research Centre	Molecular lab with PCR machines, glasshouses	BSL 1	Fully equipped	Funding gap due to limited funding agency	Specialized procurement/ waiver/ exemptions, maintenance

The main certified facilities for conducting GEd research and innovation, as well as for bioinformatics, are presented in Table 7.

3.8.2.1. Role of National Commission of Science, Academy of Scientific Research and Technology (ASRT) in Egypt

The ASRT is Egypt's main organization for planning scientific research activities. It comprises the national committees of 20 International Scientific Unions and 15 specialized Councils covering various scientific fields. The Supreme Council of Universities has exclusive authority in Egypt to recognize academic degrees, diplomas, and certificates issued by universities. Meanwhile, the Supreme Council of Institutes establishes the overall policy for institutes within the framework of higher education planning. It also approves internal regulations covering study durations, scientific content, departments, staff, student admission conditions, faculty standards, and examination procedures. The Agriculture Export Council is a nonprofit body that promotes agricultural exports, advises the Minister of Trade and Industry, and represents Egyptian exporters of agricultural crops. The Higher Council for Science and Technology (HCST), overseen by the Ministry of Scientific Research, focuses on strategic planning and the development of long-term visions for science and technology. It adopts the standards outlined in the executive plan and monitors their implementation to support the achievement of the national strategy's goals.

3.9. Traditional and Staple Crops, Livestock, Agroforestry, and Fisheries Varieties/

Breeds for Improvement Using GEd

Genome Editing (GEd) technology offers targeted solutions for Egypt's agricultural bottlenecks—specifically extreme water scarcity, soil salinity, and high heat. Egypt has a diverse selection of commercial and native crops suitable for GEd technology. Cairo University is conducting research to develop drought-resistant maize, wheat, and chickpea varieties. The most critical crops that stand to benefit from this biotechnology include wheat, maize, cotton, faba beans and sun flowers. The indigenous and commercial crops with potential for GEd research in Egypt are detailed in Table 8.

Table 8: National priority crops that could potentially benefit from GEd technology

Organism / Species	Trait improvement of Interest	Socioeconomic Justification	GEd Potential (Low/Medium/High)	Existing R&D	Actual Annual Production (tonnes)
Cereals					
Wheat (<i>Triticum aestivum</i>)	Fungal resistance, Yield	Staple crop, Central to food and nutrition security	High	Import >50% of production	9,200
Maize	Insect resistance, Yield	Food & feed Source of oil Not self-sufficient	High	Food & feed Source of oil Not self-sufficient	7,200
Barely	Draught Stress and Nutrition	Improved yield and high nutrition	Medium	Limited production	90
Sorghum	Draught stress and nutrition	Improved yield and high nutrition	Medium	Limited production	790
Legumes					
Faba beans	Broomrapes Disease resistance	Major source of protein and income	High	Source of plant protein for low-income	277.3
Lentil	Pests and Disease	Major source of protein	Medium		
Cowpea	Pests and Disease	Major source of protein and income	Medium		
Beans	Pests and Disease	Major source of protein and income	Medium		
Chickpeas	Drought	Major source of protein and income	Medium		1.5
Fruit crops					
Table grapes	Drought, disease	Exportation, Nutrient-rich food and cash crop	High	Exportation	

	resistance & yield				
Strawberry	Diseases, drought	Exportation, Nutrient-rich food and cash crop	High	Exportation	
Banana	Pests and Disease	Pest and Disease Resistance	High	--	
Date palm	Insects	Exportation, Nutrient-rich food and cash crop	High		
<i>Tuber crops</i>					
Potato	Viruses, Yield	Exportation, Critical root crop for food security			
<i>Fiber crops</i>					
Cotton	Insect/ weeds	Used for industry & top global exporters	High		425 (480-lb Bales)
<i>Oil crops</i>					
Sunflower	Fungi, Yield	A significant source of oil	Medium		50

*Priority source Egypt's Sustainable Development Strategy (Egypt's Vision 2030)

According to the International Production Assessment Division, Egypt's wheat production for 2024-2025 reached 9,200 (1000 tons), with a yield of 6.57 (T/Ha). Sorghum output was 790 (1000 tons), at a 4.09 (T/Ha) yield, while barley production hit 90 (1000 tons) with a 5.27 (T/Ha) yield. Wheat and corn are Egypt's key cereal crops for food and feed. However, due to local shortfalls, Egypt's wheat imports for the 2024/25 marketing year (July–June) are projected at 12.5 million metric tons (MMT), and corn imports at 8.0 MMT. Increasing wheat and corn production and yields via GEd is essential for Egypt's food security and stability. Legumes, known as poor man's meat, offer a rich source of plant-based protein and vital nutrients, making them a healthy, versatile alternative to beef. Using GEd to boost yield and productivity lowers costs and poverty. Pests and diseases reduce yields in fruits, potatoes, cotton, and many other crops. GEd can help develop resistant plants and boost Egypt's economy, as infected produce hampers exports. Currently, Egypt lacks GEd for livestock and fisheries; future planning should include these sectors.

3.10. Intellectual Property Rights and Benefit Sharing

The Egyptian Patent Office (EPO) aims to effectively manage the patent system to promote knowledge sharing, motivate inventors, and safeguard their rights. It achieves this through administrative improvements, business governance, and the adoption of national patent

policies that support broader societal development across all areas (Table 9). The EPO encourages innovation and invention, fostering a society capable of producing and utilizing technology efficiently to support comprehensive development. The EPO operates under the Academy of Scientific Research and Technology. In 2023, the Egyptian Authority for Intellectual Property (“IP Authority”) was established by Law No. 163 of 2023. The IP Authority has exclusive and comprehensive competence over all matters relating to Intellectual Property Rights (“IPRs”) and will be affiliated with the Egyptian Cabinet. The IP Authority has several competencies, including “Registration of IPRs and issuance of certificates of protection,” which will allow scientists to register their inventions and products, such as GEd. In addition, new plant varieties are protected under Book 4 of the IP Rights Law 82, 2002 (<https://www.egypt.gov.eg/en/legislation/laws/82-2002>). Although Egypt is involved in several GEd projects, none have yet developed products for patenting.

At the global level, Egypt is a member of several international treaties and organizations that facilitate the protection of IPR globally. These include the World Trade Organization, the World Intellectual Property Organization (WIPO), the Patent Cooperation Treaty (PCT), the Madrid Protocol for the International Registration of Marks, and the Lisbon Agreement Concerning the Protection of Appellations of Origin and their International Registration. In 2005, the Agreement on Trade-Related Aspects of Intellectual Property Rights (TRIPS) entered into force in Egypt, obligating the country to implement the agreement and to ensure full protection of intellectual property rights, including copyright, trademarks, and patents.

Table 9: IPRs, legal instruments (laws) and organizations in charge

Intellectual Property Rights	Organization / Institution	Laws / Regulations / Protocols / Treaties
	Egyptian Authority for Intellectual Property (“IP Authority”)	Law No. 163 of 2023
Patents, Trademarks, and Copyrights	Egyptian Patent Office (EPO) under Academy of Scientific Research & Technology	
Protection of IPRs (e.g.....)	World Trade Organization (WTO)	Law No. 82 of 2002
International patent applications	World Intellectual Property Organization (WIPO)	Patent Cooperation Treaty (PCT) 1970 (amended 1979, 1984, 2001)
International registration of marks		Madrid Protocol 1989, enforced 1995
Protection of Appellations of Origin and their International Registration		Lisbon Agreement 1958

3.11. Private Sector Participation

International collaboration is essential to the development of genome-editing technologies in Egypt, as it would strengthen research capacity, facilitate the exchange of ideas and knowledge, build human capacity, and advance the development of locally adapted solutions to local health and agricultural problems. Private-sector companies in Egypt support the adoption of new biotech crops by providing techniques, information, and capacity-building, including Bayer-Egypt Pvt Limited, PICO Modern Agriculture Company, and Pearl of Africa (Table 10). Research institutes and universities in Egypt collaborate with national and international organizations, such as ICARDA, USDA, ISAAA, the Alliance for Science, and Arab Centre for the Studies of Arid Zones and Dry Lands (ACSAD) the International Potato Centre (CIP) For example, a partnership among Cairo University, ICARDA, AGERI, and Tuskegee University (USA) is advancing genome-editing projects to develop drought-tolerant chickpea varieties. Also, collaborations among the Agriculture Research Centre (ARC) of Egypt, the International Institute of Tropical Agriculture (IITA), is underway to develop plant genome-editing tools. Collaborations among AGERI, Cairo University, and PICO Seeds Company to develop biotech strawberry for virus resistance and gerbera plants for flower colour were also permitted, engaging the private sector in biotechnology research. Challenges and opportunities facing engaging in the private sectors is illustrated in Table 11.

Table 10: Overview of Private Sector Companies involved in Seed Production & Distribution, Crop Improvement & Biotech/GEEd activities in Egypt

Company	Multinational / Regional / National	Biotech/GEEd Activities	Partnerships	Challenges Faced	Investment Interest
Corteva Egypt Seeds	Multinational	No Biotech/GEEd. High-tech conventional seed breeding (trait selection) & seed applied technologies, seed distribution. It drives agricultural biotechnology with advanced seed solutions and crop protection products tailored for local farmers	ARC, Agro-input dealers	No active GMO Regulations, regulatory, registration or certification	Crop breeding (Biotech/GEEd & conventional). It Incorporates science-backed biological solutions that protect crops from abiotic stress and pests while promoting sustainable growth without chemical reliance
Bayer Limited Egypt LLC	Multinational	No Biotech/ GEEd. High-tech conventional seed breeding (trait selection) & seed applied technologies, seed distribution. It provides advanced agricultural biotechnology solutions	ARC, MayTrade, Agrimatco, Samtrade, ATI Agro Egypt	No active GMO Regulations, regulatory, registration or certification	Crop breeding (Biotech & conventional) Including providing hybrid seeds, climate-resilient traits, and systemic biological and chemical crop protection against pests and foliar diseases
Syngenta	Multinational	No Biotech/GEEd. High-tech conventional seed breeding (trait selection) & seed applied technologies, seed distribution	ARC, KAHA Research Station, Lotus Agriculture Development, Al Dahra, Emerald Seed Company	No active GMO Regulations, regulatory, registration or certification	Crop breeding (conventional) such as using bio-stimulants and bio-controls enhance soil health, manage abiotic stresses, and improve nutrient absorption Biologicals

PICO Modern Agriculture Company	National & Regional	No Biotech/GEEd	Cairo University, AGERI, EAGA, AEC, Commercial growers	Lack of CRISPR tools. Strict GMO Regulations, regulatory, registration & certification bottlenecks	Crop breeding (conventional) Biotechnology (strawberry & gerbera)
Fine Seeds International	National & Regional	Biotech/No GEEd.	Cairo University, AGERI, Bayer, Bejo Zaden, US Agri Seed, Seminis, KWS, VoloAgri Group	Lack of CRISPR tools, Strict GMO Regulations, regulatory, registration & certification bottlenecks	Crop breeding (conventional) Previous work on GM maize
Pearl of Africa	Continental	Biotech/No GEEd	Cairo University, ARC, DRC, Gaara Seeds, Nile Corporation for Agricultural Development, MK Group, MALR, NECSDC, DRC	Lack of CRISPR tools, Strict GMO Regulations, regulatory, registration & certification bottlenecks	Animal breeding (conventional)
ACSAD	Regional (Arab countries)	Biotech/GEEd	ARC/AGERI, CEEBA, Beltone Holding, National Seeds Co	Lack of CRISPR tools, Strict GMO Regulations, regulatory, registration & certification bottlenecks	Crop/ Animal breeding (conventional)
ICARDA	International (CGIAR)	Biotech/GEEd	Cairo University, ARC, CASP, CASC	No active GMO Regulations, regulatory, registration or certification	Crop (Biotech/GEEd & conventional) Animal (conventional)

Table 11: Challenges and opportunities of engaging in the private sectors

Challenges	Opportunities
Ethical Concerns of genetic modification—particularly in human germline editing—and the broader ecological and socioeconomic concerns surrounding corporate agriculture make public buy-in a hurdle.	Human capacity and highly skilled researchers
Lack of Awareness & Public Acceptance	Availability of International Research Programmes
Limited expertise in GEd techniques	Political willingness to demand Climate-Resilient Crops and to feed the growing population
Limited Investment in GEd R&D and limited access to funds	Public-Private Partnerships (PPPs): Private companies can partner with academic powerhouses to translate laboratory CRISPR/Cas9 innovations into marketable products.
Market Access and Readiness: The high cost of genetic/genomic testing and gene therapies makes these products inaccessible to large segments of the Egyptian population.	Regional Hub Potential: Egypt has the capability and infrastructure to act as a primary catalyst and hub for genetic and biotechnology applications across the Middle East and North Africa
Regulatory Uncertainty: Egypt lacks a finalized, comprehensive biosafety and legal framework specifically addressing genome editing.	

3.12. Funding and Investment Landscape

The following are the leading institutions supporting research in Egypt:

Academy of Scientific Research and Technology (ASRT): ASRT serves as the executive arm of the Egyptian Ministry of Scientific Research, responsible for designing and implementing national programmes and roadmaps. It hosts sectoral councils on medical research and ethics, strategic committees such as the national pharmacology committee, key departments including the patent office and the National Innovation and Invention Development Agency, and a scientific observatory. ASRT serves as Egypt's national house of expertise and national think tank in Science, Technology, and Innovation, focusing on STI strategic planning, preparing technological roadmaps and future studies, recognizing excellence in science, benchmarking research institutions, and raising scientific awareness. ASRT provides consultations to the government and decision-makers on science and technology issues in Egypt and worldwide—the primary public supporters of Innovation, Entrepreneurship, IPR, Technological Incubators, and Technology transfer.

Science and Technology Development Fund (STDF): STDF is the major research-funding organization and continuously improves its performance. STDF has stimulated the Egyptian scientific community by funding distinguished research papers and establishing scientific partnerships with scientists from many advanced countries to keep pace with rapidly advancing technology, be open to diverse cultures and new economic unions, compete on the international stage, link scientific research to technological development, and cooperate with civil society institutions to activate their role in the integrated scientific research system. STDF's main objectives are to improve the research and development environment; provide funding for scientific research and technology development; support and develop the innovation capabilities of the science and technology community; support the full cycle of scientific research and product development; and disseminate information on science and technology in Egypt. USAID-STDF offers research grants for collaborations between Egyptian and U.S. scientists and establishes Centres of Excellence in water, energy, and agriculture. STDF funded 4 GEd projects (wheat, chickpea, maize, and sorghum) totalling \$800,000.

Research Institutions and Universities: As discussed above, each university and research institute has its own funds to promote and support research and to fund graduate theses. They invest in knowledge by providing infrastructure, training, and workshops.

In Egypt, funding for genome editing research and development is mainly derived from government allocations. An overview of national funding sources for genome editing projects/programs is presented in Table 12.

Table 12: Overview of National and Other Funding Sources for Genome Editing

Funder/Donor	Organisation Type	Genome Editing Project	Amount (USD)	Duration	Recipient Institution(s)	Area of Focus
Science and Technology Development Funds (STDF)	Government R&D Funding Agency	- National - Collaborative Research & Innovation Grants	- 100-150 per project - 200,000 per partner	2–3 years	Cairo University, AGERI	Drought tolerance, Yield improvement, Biofuel Production and Forage Quality
Academy of Scientific Research and Technology (ASRT):	Government Ministry	National Biotechnology Network of Expertise (NBNE)	Annual budget allocation (~300,000+ to R&D)	Ongoing	Led by Cairo University with other 18 Universities and Institutes	Communication, technology transfer, build network
Egypt Genome Project	Government Ministry	Human Genome	19 million	Ongoing	Egypt Centre for Research and Regenerative Medicine (ECRRM)	Genome study, Bioinformatics, Nutrient improvement

4. RECOMMENDATIONS AND POLICY OPTIONS

Signatory and ratification of multilateral environmental treaties and agreements

Egypt is a signatory to numerous multilateral environmental and food agreements and treaties, including the CAC, UNFCCC, the CBD, the Cartagena Protocol, and the Nagoya-Kuala Lumpur Supplementary Protocol on Liability and Redress, Access and Benefit Sharing. Egypt has demonstrated a strong commitment to global environmental and sustainability efforts, aligning its national strategies with international goals to combat climate change, protect biodiversity, ensure food safety, and preserve the ozone layer. This commitment is underscored by hosting COP27, launching the 2050 National Climate Change Strategy, and participating in various UN-backed initiatives.

Policy implications/options

Egypt aligns its food safety and environmental policies with international standards by integrating global treaties into its national framework, notably through the National Food Safety Authority (NFSA) and the 2050 Climate Change Strategy. Key measures include adherence to Codex Alimentarius food standards, fulfilment of Paris Agreement commitments (NDC updates), and implementation of environmental compliance monitoring through the Egyptian Environmental Affairs Agency (EEAA).

Regulatory and policy frameworks

Egypt lacks a dedicated, formal regulatory framework for genome editing technologies; existing biotechnology regulations primarily cover Genetically Modified Organisms (GMOs). While guidelines exist to classify GE products by final characteristics (non-transgenic vs. transgenic), there is no dedicated, comprehensive policy governing the research, production, or importation of gene-edited animals or crops, creating a regulatory gap.

Capacity building in GEd technologies

Egypt has made strides in genomic research, notably through the Egyptian Genome Project, but still faces significant capacity-building gaps in gene editing. Key challenges include limited funding, a lack of specialized training, inadequate infrastructure for advanced molecular biology, and regulatory bottlenecks. Capacity building in genome editing technology is critical to enabling responsible, safe, and effective deployment, particularly in sectors such as agriculture and medicine. A comprehensive approach that combines theoretical knowledge with hands-on training for both scientists and regulators is essential to keep pace with rapid innovation and ensure proper oversight.

Communication and Public Engagement

Implementing robust, innovative public education on gene editing in Egypt requires a multi-stakeholder strategy that engages media, policymakers, and local communities to balance risks and benefits. Strategies include developing tailored, transparent, and linguistically accessible campaigns that address cultural context and leverage partnerships. Key approaches for a comprehensive campaign include creating specialized content for scientists and policymakers on regulatory frameworks and simplified, trust-building communication for the public to address misconceptions. Involve local communities and indigenous companies in dialogues to ensure that GEd applications (e.g., in agriculture) align with local needs and values. Utilize UNDP-style training modules to foster critical thinking, helping the public navigate complex scientific information and potential misinformation surrounding GEd risks, and to facilitate dialogue on "agile" governance models among politicians and experts to manage potential socioeconomic impacts proactively. This initiative should leverage Egypt's growing, innovative GEd Tech sector to create interactive platforms that ensure the campaign is engaging, accessible, and inclusive, thereby building informed consent and public trust.

Fostering international and regional collaborations

Egypt's strategic focus on international and regional partnerships is a cornerstone of its capacity in gene editing and biotechnology. These collaborations are designed to address critical national goals, including food security, public health, and economic resilience; harmonize regulations; attract and secure funding; and ensure equitable access to GEd technology. This would accelerate the responsible adoption, diffusion, and realization of the benefits of GEd technology in addressing local and continental challenges. Collaboration with regional bodies is crucial for Egypt to develop a predictable and harmonized regulatory framework for genome-edited products. Active participation in initiatives such as AUDA-NEPAD and partnerships with bodies enable Egypt to align with continental standards, thereby easing regulatory hurdles for, for example, agricultural biotechnology. Egypt is actively strengthening partnerships with international research centres, academic institutions, and the private sector to foster knowledge transfer, technical assistance, and innovation, in line with Egypt Vision 2030. Key focus areas include digital transformation, higher education, AI development, and technical vocational training, with significant investment in PPP models to enhance infrastructure and human capital.

Investment in GEd R&D strategies in Egypt

Egypt is actively investing in gene editing for agricultural enhancement, focusing on developing drought-tolerant wheat, chickpeas, and improved sorghum to boost food security.

These efforts include capacity-building at the Agricultural Research Centre (ARC) and at Cairo University to establish regulatory frameworks that distinguish GEd products from traditional GMOs. Efforts to advance CRISPR/Cas9 technology are increasingly supported by targeted capacity-building, including the establishment of new regulatory frameworks that differentiate gene-edited products from conventional GMOs. While funding can be limited in certain regions, investments are being channelled into infrastructure and training initiatives to transfer gene-editing expertise to local researchers and institutions. Key strategies for the responsible and effective development of gene-editing capabilities include providing advanced infrastructure and training for scientists and regulators, promoting regional and international collaboration, fostering public trust through transparent communication, and developing flexible, harmonized biosafety regulatory frameworks.

Summarized matrix of action plans according to: Issues → gaps → recommended actions → lead institutions.

Egypt actively implements GEd research projects but lacks genome-editing policies. It is essential to develop product-based biosafety guidelines to enhance the development of GEd products. In addition, recommendations include continued development of robust biosafety frameworks, capacity building for decision-makers, stakeholder engagement, and fostering an environment that supports local researchers and businesses in leveraging genome editing for food security and economic growth.

According to the assessment review for Egypt, the significant recommendations of this report are summarized as follows:

- 1. Establish inter-ministerial/Departmental coordination or National Biosafety Authority (NBA) bodies for GEd governance and policy coherences**
- 2. Establish a clear and inclusive legal framework for GEd** by developing a national regulatory framework for genome editing, aligned with international instruments and involving key stakeholders in regulatory design to enhance legitimacy and applicability.
- 3. Strengthen institutional coordination** by building a database for GEd research projects, capacity-building initiatives, and expert scientists, and by encouraging joint projects between research institutes nationally and internationally.
- 4. Invest in capacity building and infrastructure** by supporting training programmes in GEd, bioinformatics, and regulatory science for researchers, technicians, and regulators, and by enhancing infrastructure by providing the required equipment and facilities to enable advanced genome editing and detection technologies.
- 5. Fund strategic GEd project and programmes** targeting national challenges.

- 6. Promote private sector participation** by initiating startup incubation programmes in agricultural biotechnology and genome editing, as well as by starting biotech companies targeting public-private partnerships
- 7. Develop guidelines for ethical and responsible research**
- 8. Raise public awareness and trust** by developing communication strategies on the benefits of genome editing engaging farmers, consumers, and civil society
- 9. Mobilize sustainable funding mechanisms** by increasing allocated funds for GEd at the national level, as well as mobilizing external funds.
- 10. Include research on GEd for livestock and fisheries** to improve their productivity and improve their resistance to diseases.

REFERENCES

- Abdallah, N. A. (2010). GM crops in Africa: Challenges in Egypt. *GM Crops*, 1(3), 116–119.
<https://doi.org/10.4161/gmcr.1.3.12811>
- Abdallah, N. A. (2025). Breeding plants resilient to climate change. In M. T. Khalil, W. W. M. Emam, & A. Negm (Eds.), *Climate change impacts on aquatic environment* (Earth and Environmental Sciences Library). Springer. https://doi.org/10.1007/978-3-031-74897-4_8
- Abdallah, N. A., Elshaday, H., Abulela, H. A., Thilmony, R., Abdelhadi, A. A., & Elarabi, N. I. (2022). Multiplex CRISPR/Cas9-mediated genome editing to address drought tolerance in wheat. *GM Crops & Food*, 16(1), 1–17.
<https://doi.org/10.1080/21645698.2022.2120313>
- Abdallah, N. A., Hamwieh, A., Baum, M. (2024). Novel genome-editing approaches for developing non-GM crops for sustainable improvement and the mitigation of climate change. In J. T. Chen & S. Ahmar (Eds.), *Plant genome editing technologies* (Interdisciplinary Biotechnological Advances). Springer. https://doi.org/10.1007/978-981-99-9338-3_3
- Abdallah, N. A., Hamwieh, A., Radwan, K., Fouad, N., & Baum, M. (2023). Genome editing and global food security: Molecular engineering technologies for sustainable agriculture. In *Molecular engineering technologies for sustainable agriculture*. CRC Press. <https://doi.org/10.4324/9781003382102-5>
- Abdallah, N. A., Hamwieh, A., Radwan, K., & Baum, M. (2024). Strategies for speed breeding in crops to accelerate plant improvement. In *Plant speed breeding and high-throughput technologies*. CRC Press. <https://doi.org/10.1201/b23372>
- Abdallah, N., Sabouni, K. A., Hamwieh, A., Abdallah, A., Alsekhni, N., Elarabi, N., Radwan, K., Veeramasu, S., & Prakash, C. (2025). Agrobacterium rhizogenes-mediated CRISPR/Cas9 base editing of the CaSal1 gene enhances drought tolerance in chickpea. *International Journal of Agriculture & Biology*, 33, 330210.
<https://doi.org/10.17957/IJAB/15.2270>
- Abdelnour, S. A., Xie, L., Hassanin, A. A., Zuo, E., & Lu, Y. (2021). The potential of CRISPR/Cas9 gene editing as a treatment strategy for inherited diseases. *Frontiers in Cell and Developmental Biology*, 9, 699597.
<https://doi.org/10.3389/fcell.2021.699597>
- Abdulla, M. F., Mostafa, K., & Kavas, M. (2024). CRISPR/Cas9-mediated mutagenesis of

- FT/TFL1 in petunia improves plant architecture and early flowering. *Plant Molecular Biology*, 114, Article 69. <https://doi.org/10.1007/s11103-024-01454-9>
- Ahmed, A., Ashraf, D., Bahaa, A., El-Tayebi, H., & Adwan, H. (2020). Impact of CDK4 knock out using CRISPR/Cas9 gene editing technology on breast cancer progression. *European Journal of Cancer*, 138, S70–S71.
- Alam, M. S., Kong, J., Tao, R., Ahmed, T., Alamin, M., Alotaibi, S. S., Abdelsalam, N. R., & Xu, J.-H. (2022). CRISPR/Cas9-mediated knockout of the OsbHLH024 transcription factor improves salt stress resistance in rice (*Oryza sativa* L.). *Plants*, 11, 1184. <https://doi.org/10.3390/plants11091184>
- Alkompoz, A. K., Hamed, S. M., Zaid, A. S. A., Almangour, T. A., Al-Agamy, M. H., & Aboshanab, K. M. (2023). Correlation of CRISPR/Cas and antimicrobial resistance in *Klebsiella pneumoniae* clinical isolates recovered from patients in Egypt compared to global strains. *Microorganisms*, 11(8), 1948. <https://doi.org/10.3390/microorganisms11081948>
- Ali, N. M., Abdel-Salam, M., Hanfy, A., Abd-Elhalim, H. M., Ahmed, N. E., & Metry, E. A. (2023). Increasing amylopectin in the tetraploid potato Desiree cultivar (*Solanum tuberosum* L.) via CRISPR/Cas9. *Arab Universities Journal of Agricultural Sciences*, 31(2), 173–184. <https://doi.org/10.21608/ajs.2023.203532.1518>
- Amiri, F., Ranjbar, M., Pirouzfard, M., et al. (2021). HLA-A gene knockout using the CRISPR/Cas9 system to overcome transplantation concerns. *Egyptian Journal of Medical Human Genetics*, 22, 37. <https://doi.org/10.1186/s43042-021-00155-y>
- Assem, S. K., Basry, M. A., Taha, T. A., El-Aziz, M. H. A., Alwa, T., & Fouad, W. M. (2023). Development of an in vitro regeneration system from immature inflorescences and CRISPR/Cas9-mediated gene editing in sudangrass. *Journal of Genetic Engineering and Biotechnology*, 21(1), 58. <https://doi.org/10.1186/s43141-023-00517-6>
- Cartagena Protocol on Biosafety. (2000). *Cartagena protocol on biosafety to the convention on biological diversity*. United Nations Environment Programme. <https://bch.cbd.int/protocol/>
- Duan, X., Ma, W., Jiao, Z., et al. (2022). Reverse transcription-recombinase-aided amplification and CRISPR/Cas12a-based visual detection of maize chlorotic mottle virus. *Phytopathology Research*, 4, 23. <https://doi.org/10.1186/s42483-022-00128-y>
- European Commission/CORDIS. (2013). *Boosting Tunisian biotech research*. <https://cordis.europa.eu/article/id/90116-boosting-tunisian-biotech-research>
- Hassan, S. M., Abdel-Salam, M. A., Hanfy, A., Abd-Elhalim, H. M., Ahmed, N., & Metry, E. A. (2023). Decreasing monosaccharide contents resulted from cold-induced sweetening in tetraploid potato (*Solanum tuberosum* L.) using genome editing. *Arab*

- Universities Journal of Agricultural Sciences*, 31(2), 157–171.
<https://doi.org/10.21608/ajs.2023.198877.1517>
- International Service for the Acquisition of Agri-biotech Applications. (2020). *Global status of commercialized biotech/GM crops: 2020* (ISAAA Brief No. 55).
<https://www.isaaa.org>
- Madkour, M. A., El Nawawy, A. S., & Traynor, P. L. (2000). *Analysis of a national biosafety system: Regulatory policies and procedures in Egypt* (ISNAR Country Report No. 62). International Service for National Agricultural Research.
https://www.researchgate.net/publication/237614295_Analysis_of_a_National_Biosafety_System_Regulatory_Policies_and_Procedures_in_Egypt
- Mohr, T., Horstman, J., Gu, Y. Q., Elarabi, N. I., Abdallah, N. A., & Thilmony, R. (2022). CRISPR-Cas9 gene editing of the Sal1 gene family in wheat. *Plants*, 11, 2259.
<https://doi.org/10.3390/plants11172259>
- Rahim, A. A., Uzair, M., Rehman, N., Fiaz, S., Attia, K. A., Abushady, A. M., Yang, S. H., & Khan, M. R. (2024). CRISPR/Cas9-mediated TaRPK1 root architecture gene mutagenesis confers enhanced wheat yield. *Journal of King Saud University – Science*, 36(2), 103063. <https://doi.org/10.1016/j.jksus.2023.103063>
- Secgin, Z., Uluisik, S., Yildirim, K., Abdulla, M. F., Mostafa, K., & Kavas, M. (2022). Genome-wide identification of the aconitase gene family in tomato (*Solanum lycopersicum*) and CRISPR-based functional characterization of SIACO2 on male sterility. *International Journal of Molecular Sciences*, 23(22), 13963.
<https://doi.org/10.3390/ijms232213963>
- Tang, Y., Abdelrahman, M., Li, J., Wang, F., Ji, Z., Qi, H., Wang, C., & Zhao, K. (2021). CRISPR/Cas9 induces exon skipping, facilitating the development of fragrant rice. *Plant Biotechnology Journal*, 19(4), 642–644. <https://doi.org/10.1111/pbi.13514>
- USAID. (2020). *Biotechnology and agricultural innovation: A review of development and funding programs*. USAID. <https://www.usaid.gov>
- Wei, Z., Abdelrahman, M., Gao, Y., Ji, Z., Mishra, R., Sun, H., Sui, Y., Wu, C., Wang, C., & Zhao, K. (2021). Engineering broad-spectrum resistance to bacterial blight by CRISPR-Cas9-mediated precise homology directed repair in rice. *Molecular Plant*, 14(8), 1215–1218. <https://doi.org/10.1016/j.molp.2021.05.012>
- Wanzhi, Y., Hossain, R., Pröbsting, M., Ali, A. A. M., Han, L., Miao, Y., Rietz, S., Cai, D., & Schenke, D. (2024). Knock-out of BnHva22c reduces the susceptibility of *Brassica napus* to infection with the fungal pathogen *Verticillium longisporum*. *The Crop Journal*, 12(2), 503–514. <https://doi.org/10.1016/j.cj.2024.02.012>

ANNEXES.

Annex 1. List of institutions and resource persons involved in the interview

SN	NAME	SECTOR	MINISTRY/ DEPARTMENT/ INSTITUTION/ ORGANIZATION	POSITION	CONTACT DETAILS
1	Prof. Saad Nassar	Government Departments	Ministry of Agriculture and Land Reclamation	Minister Advisor	saad.nassar@yahoo.com
2	Prof. Adel El-Sayed El- Tarras	Government Departments	Ministry of Agriculture and Land Reclamation	Minister Advisor	adeltarras@hotmail.com
3	Dr. Shireen Assem	Government Departments	Agricultural Research Centre	Vice president	shireen_assem@yahoo.com, shireenassem@ageri.sci.eg
4	Dr. Hanaiya El Itriby	Government Departments	Agriculture development projects, Ministry of Agriculture and Land Reclamation	Deputy director	elitriby.hanaiya@gmail.com
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